DIGITAL ZERO-NOISE EXTRAPOLATION FOR QUANTUM ERROR MITIGATION

Introduction

Quantum computing in the NISQ (Noisy Intermediate-Scale Quantum) era faces several challenges, including gate errors, decoherence, calibration uncertainty, and hardware limitations. These noise sources can reduce the accuracy and reliability of quantum computation results. Quantum Error Mitigation (QEM) provides practical techniques to improve computational results without requiring fully fault-tolerant quantum computers. This project focuses on **Digital Zero-Noise Extrapolation (ZNE)**, a technique designed to reduce the impact of noise in quantum computations.

The main problem addressed by this project is the loss of accuracy caused by noise in quantum circuits. Digital ZNE approaches this problem by executing a quantum circuit at different noise levels and extrapolating the measured results toward the zero-noise limit. The project uses **unitary folding** to intentionally increase the noise level through logically canceling operations, allowing the effect of noise to be studied and reduced. Different extrapolation techniques, including linear, polynomial, Richardson, exponential, and adaptive methods, will be compared using small randomized and variational quantum circuits.

The proposed system will evaluate the effectiveness of these techniques by comparing unmitigated and mitigated results using measures such as absolute error, expectation-value deviation, error reduction, gate count, and number of circuit executions. Ideal or low-noise results will be used as reference values. The expected outcome is to identify suitable ZNE methods for practical quantum error reduction while understanding the trade-off between improved accuracy and computational overhead. This can contribute to improving the quality and reliability of quantum computations on current noisy quantum hardware.

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